MACHINE LEARNING • END-TO-END PROJECT

Customer Purchase Prediction

Using Logistic Regression & Random Forest

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Faculty Project Presentation | Dataset: 5,000 records

Project at a glance

5,000
Original customer records

13
Original dataset columns

4,970
Records after cleaning

11
Predictive features

THE CORE QUESTION

Can a customer’s profile and engagement behavior help predict whether they will make a purchase?

Problem statement & objectives

Business problem

Customer behavior varies across visits, purchases, email engagement and cart activity. A structured model can estimate purchase propensity.

Project objectives

Clean and explore data

Build binary classification models

Compare models on held-out test data

Explore segments using clustering

Understanding the dataset

9 numerical predictors

Age • Annual_Income • Previous_Purchases

Website_Visits • Pages_Viewed

Time_on_Website_Min • Emails_Opened

Cart_Additions • Days_Since_Last_Purchase

2 categorical predictors

Gender • Customer_Segment

TARGET: Purchase

0 = No Purchase | 1 = Purchase

Customer_ID is an identifier, not a predictor.

BINARY CLASSIFICATION

Data quality & cleaning

30
Duplicate records removed

45
Missing cells identified

4,970
Final unique records

994
Held-out test records

PREPROCESSING STRATEGY

Numerical: median imputation + StandardScaler | Categorical: mode imputation + OneHotEncoder | Drop Customer_ID

Data leakage prevention

01 Remove duplicates → **02** Stratified 80:20 split → **03** Fit preprocessing on train → **04** Transform test with fitted pipeline → **05** Evaluate unseen test data

Train: 3,976 records • Test: 994 records • Stratified split with random_state = 42

Exploratory data analysis

EDA performed

Descriptive statistics

Target class distribution

Numerical distribution plots

Boxplots for outlier inspection

Correlation heatmap

Key observed signals

Cart_Additions: ~ +0.18

Days_Since_Last_Purchase: ~ −0.18

Emails_Opened: ~ +0.13

Annual_Income: ~ +0.12

Overall linear associations were modest.

Class distribution after cleaning



Class imbalance is moderate; stratification preserves the target ratio.

Complete ML workflow

RAW DATA

5,000 × 13

CLEAN

4,970 rows

EDA

Inspect patterns

PIPELINE

Impute • Encode • Scale

SUPERVISED

LogReg • Random Forest

EVALUATE

Accuracy • F1 • ROC-AUC

Unsupervised exploration: K-Means • DBSCAN • Hierarchical clustering (separate analytical branch)

Model 1 | Logistic Regression

WHY THIS MODEL?

Interpretable linear baseline for binary classification

Uses sigmoid function to estimate purchase probability

Default decision threshold: 0.50

Numerical scaling is important for stable fitting

61.77%

Test accuracy

0.6351

ROC-AUC

Model 2 | Random Forest

WHY THIS MODEL?

Ensemble of decision trees

Can capture nonlinear feature interactions

Baseline: 100 trees (random_state = 42)

Tuned with GridSearchCV using 5-fold CV

57.95%

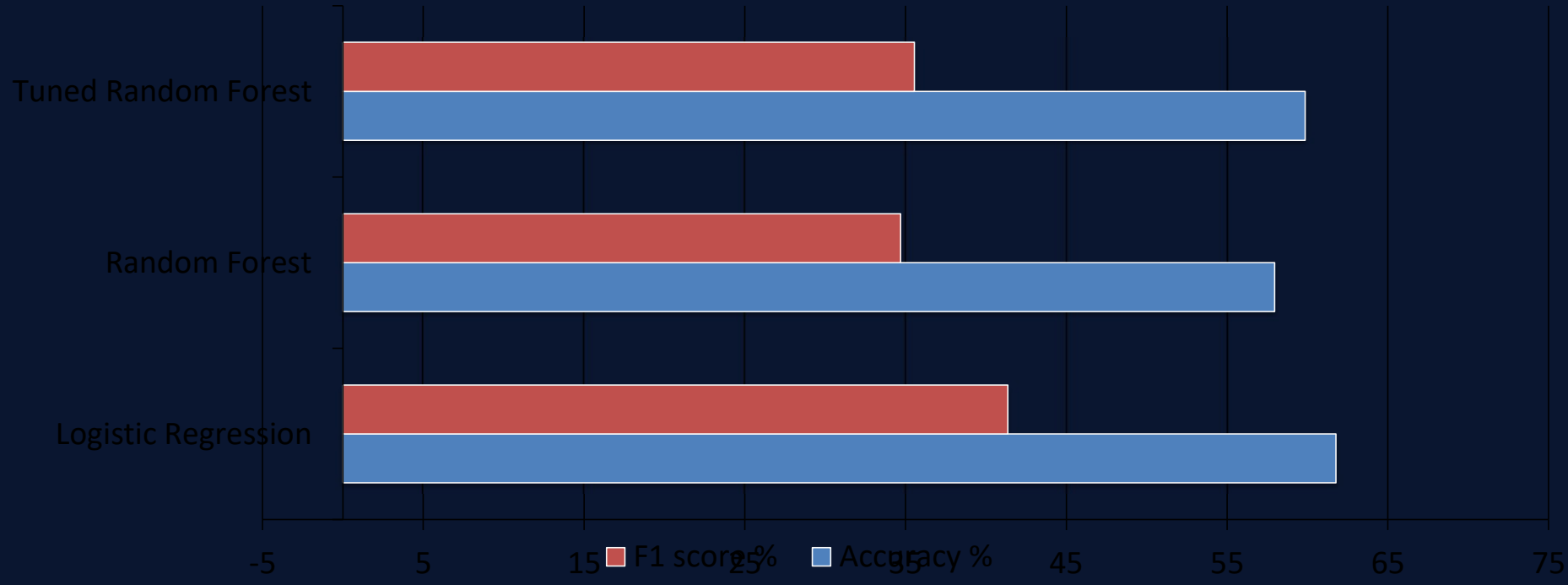
Baseline test accuracy

59.86%

Tuned test accuracy

CUSTOMER PURCHASE PREDICTION

Model performance comparison



Logistic Regression achieved the strongest reported test accuracy and F1 score.

Evaluation | All key metrics

MODEL	ACCURACY	PRECISION	RECALL	F1	ROC-AUC
Logistic Regression	61.77%	57.02%	32.45%	41.36%	0.6351
Random Forest	57.95%	48.90%	26.88%	34.69%	0.5775
Tuned Random Forest	59.86%	53.40%	26.63%	35.54%	0.6046

Purchase-class recall remains low, so the models miss many actual buyers.

CUSTOMER PURCHASE PREDICTION

Confusion matrix | Logistic Regression



Hyperparameter tuning

SEARCH METHOD

GridSearchCV with 5-fold cross-validation

Model: Random Forest

Scoring used for best CV result: accuracy

Best CV accuracy: 60.89%

BEST REPORTED SETTINGS

n_estimators = 50

max_depth = 10

min_samples_split = 5

Held-out test accuracy: 59.86%

Unsupervised learning | Separate branch

K-MEANS & HIERARCHICAL

Elbow Method for selecting K

Silhouette Score to assess separation

Best tested silhouette: ~0.0837 at K = 2

Ward linkage and dendrogram for hierarchy

DBSCAN

Density-based clustering

Parameters: eps = 1.5, min_samples = 5

Reported 32 clusters

Reported 1,353 noise points

Clustering is exploratory, not a predictor.

CUSTOMER PURCHASE PREDICTION

Sample customer | Prediction demo

INPUT PROFILE

Age: 35 | Gender: Male | Income: 80,000

Segment: Regular | Previous purchases: 5

Website visits: 10 | Pages viewed: 6

Time on website: 12 min | Emails opened: 4

Cart additions: 2 | Days since purchase: 60

PREDICTED CLASS

NO PURCHASE

35.04%

Reported purchase probability

Conclusion & future scope

WHAT WE ACHIEVED

- End-to-end supervised learning pipeline
- EDA and leakage-aware preprocessing
- Baseline and tuned model comparison
- Independent clustering exploration
- Logistic Regression performed best on reported metrics

NEXT IMPROVEMENTS

- Improve purchase-class recall
- Test class weighting and threshold calibration
- Evaluate on larger real-world datasets
- Monitor data drift and probability calibration
- Deploy as a customer propensity service

QUESTIONS & DISCUSSION

Thank you

Customer Purchase Prediction

Logistic Regression • Random Forest • Unsupervised Learning

Questions?